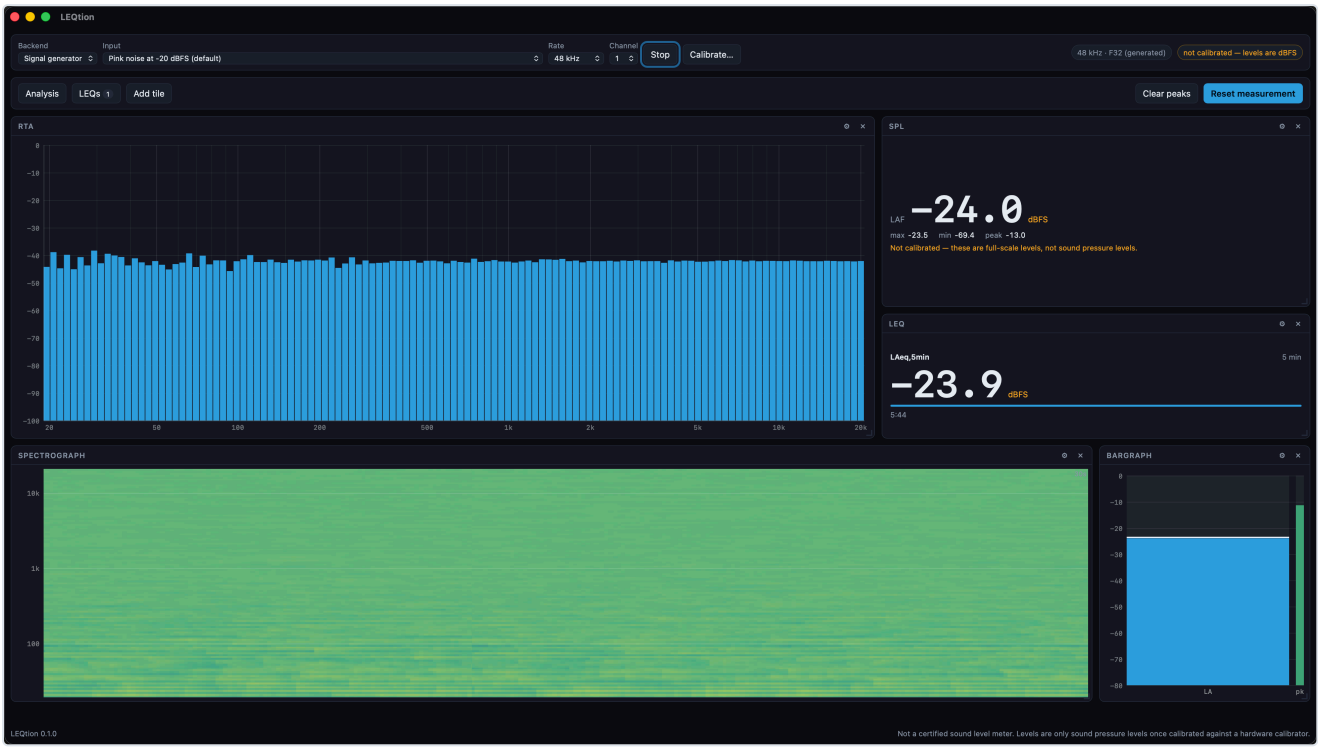
LEQtion user guide

LEQtion is a desktop sound level meter and dual-channel analyser. It has an RTA, spectrograph, bargraph, time-weighted SPL, as many user-defined LEQs as you want, level history, CSV logging, a signal generator and transfer function measurement with phase and coherence, on a grid of tiles you lay out yourself.

> **Before you rely on this:** the DSP is verified numerically and pinned as tests. The A and C weighting curves reproduce the IEC 61672-1 table to better than 0.06 dB at the midband frequencies, a full-scale sine reads 0 dBFS through the band integrator, and LEQ is checked against signals whose answer is known in advance. The audio path has been run against real Core Audio hardware on macOS.
>
> But it has **not been checked against a reference sound level meter**, **no hardware calibrator has ever been connected to it**, and **the ASIO path compiles but has never carried audio**. **LEQtion is not a certified sound level meter and makes no conformance claim.** Status: alpha.
>
> This codebase was created with AI assistance, directed and reviewed by a human author.

Start here: measure the meter, not the room

Before you plug anything in, open the **Signal generator** backend. It feeds the analyser directly, with no interface, no microphone and no microphone permission needed, and it's how you check that the meter itself is behaving.



Pink noise at −20 dBFS through the generator backend, with no hardware in the chain. Pink noise is flat per octave, so a fractional-octave RTA reading flat across the band is the check. A wrong window, a wrong normalisation or a wrong band integration would show up as a tilt or a step instead. Here LAF and LAeq,5min agree within 0.1 dB over a complete window, and every level says dBFS because nothing is calibrated.

Calibration is refused while the generator backend is open. There's no capsule in the chain, and an offset taken from a synthetic sine would be a number made up out of nothing.

The tiles

Add, remove, drag and resize them, and the layout sticks.

Tile	What it shows
RTA	Fractional-octave spectrum, 1/1 down to 1/48, with selectable transform size, window, overlap, averaging and peak hold. A strip across the top names the tallest band and its level, refined to a fraction of a bin when a tone is what tops it
Spectrograph	The same bands over time, scrolling, on the same log axis as the RTA so the two line up when stacked
Bargraph	Level meter with a held maximum, plus a separate input-peak strip that stays in dBFS because headroom is an electrical question
SPL	Time-weighted level — Fast, Slow and Impulse — with max, min and peak
LEQ	As many as you like, each with its own window and weighting
Level history	Any of those levels over time, as a line
Transfer function	Magnitude, phase and coherence against a reference

LEQs run in the engine rather than in the tile, so an LEQ keeps integrating whether or not a tile is showing it. You can close one and come back to it later.

Each LEQ takes any window length you type (or a preset from 1 second to an hour, or "since reset") and its own **weighting** (A, C or Z).

Level history and logging

Each point on a history chart covers a whole interval, and the band around the line is the min and max inside that interval, so a transient between ticks makes it onto the chart instead of slipping through. Zooming out buckets the points in the engine and keeps the extremes, so the trace never flattens as you look further back.

Data logging writes the measurement to CSV, one row per interval, every series at once. The rows are the chart's own points rather than a second sampling on a different clock.

> Every row states **whether it is calibrated** and **how many frames have been dropped**. A log that covers a gap in the audio says so in the file.

Calibration

Calibration is against a hardware acoustic calibrator, 94 or 114 dB at 1 kHz.

Until you calibrate, every level is a full-scale level and the app says so, on the SPL tile, on the LEQ tile and in the device bar. An uncalibrated number dressed up as a sound pressure level is the single most damaging thing a meter can do.

A calibration is trusted silently for the rest of a measurement, so a run has to pass four checks before it's accepted, and **there is no override**:

1. A steady level, with a spread under 0.5 dB.
2. The right frequency: the tone within 5% of what the calibrator claims.
3. No clipping.
4. A tone well clear of the noise floor.

Each refusal tells you what to do about it. "Unstable" usually means the calibrator isn't seated on the capsule.

> **What it can't check:** that the calibrator is itself in calibration, or that nobody has changed the preamp gain afterwards. **Changing gain by one click invalidates the offset completely** and nothing in software can detect it, which is why the calibration records the device it was taken on.

The signal generator

Pink noise, white noise, sine and a repeating log sweep, out of a channel you choose, with optional band-limiting.

Level is **dBFS RMS**, and the expected **peak** is shown beside it, because pink noise at −6 dBFS RMS clips hard while looking like a nice conservative setting.

Transfer function

`H = Sxy/Sxx`, the H1 estimator, from **complex-averaged** cross-spectra, against a reference that's either the generator's own output tapped internally or a hardware loopback on an input.

Coherence is drawn, not hidden. Every point is faded in proportion to it, and the magnitude trace **breaks** wherever coherence falls below the floor.

> A transfer function without coherence looks just as confident where the measurement is solid as where it's picking up the air conditioning, and people tune systems off the second kind.

Delay finding locates the arrival from the impulse response, sub-sample interpolated, and reports it in milliseconds, metres and samples with a confidence figure. The reference has to be delay-

compensated before any of the rest means anything.

Several transforms run in parallel, each serving a couple of octaves and halving in length as frequency rises, stitched onto one set of points. One FFT length can't serve both 20 Hz and 16 kHz: at 48 kHz a 16384-point transform gives 2.9 Hz bins, about right at 30 Hz and absurdly narrow at 10 kHz.

Sample rate matters more than you'd expect

The analogue A-weighting design has two more poles than zeros, and every s→z transform has to do something with that surplus. Both textbook answers are badly wrong at the top of the band:

Design	A-weighting error at 19 kHz, 48 kHz
Bilinear — forces a double zero onto Nyquist	−13.6 dB
Plain matched-Z — invents no surplus zero	+7.2 dB
What LEQtion does	**under 1 dB**

Measured worst deviation over 20 Hz – 20 kHz:

Sample rate	A	C
44.1 kHz	1.15 dB	0.30 dB
48 kHz	0.99 dB	0.21 dB
96 kHz	0.22 dB	0.01 dB

> **Run at 96 kHz if you care.** It costs nothing but CPU and makes the weighting effectively exact, which is why the app shows the sample rate rather than hiding it.

In practice the cost is smaller than the headline: on a nasty 29-tone test signal with as much energy at 16 kHz as at 1 kHz, the A-weighted level is 0.23 dB out at 48 kHz.

Dropped audio

If the analysis thread falls behind, the audio callback throws samples away rather than blocking the driver. Time then goes missing, and **every LEQ on screen is short by an unknown amount**.

The device bar says so and tells you to restart the measurement, rather than carrying on and hoping nobody notices.

Backends

Core Audio, WASAPI, ALSA and JACK out of the box. ASIO is behind a build flag and compiles, but **it has never carried audio**; see asio.md.

Command-line diagnostics

Two ship with it, for when a measurement looks wrong and the GUI is in the way.

Check that an input delivers audio at all:

```
cargo run -p leqtion-audio --example capture -- --list
cargo run -p leqtion-audio --example capture -- --seconds 3
```

> If a run reports frames arriving at **perfect digital silence**, that's macOS denying microphone access.

The whole measurement chain without the window, with the same engine and the same numbers:

```
cargo run --example meter -- --seconds 30 --offset 120
```

Troubleshooting

Symptom	Cause
Everything reads dBFS	Not calibrated. That is the design — it will not present a full-scale level as an SPL.
Calibration refused	One of the four checks failed, and the message says which. "Unstable" usually means the calibrator isn't seated.
Calibration option greyed out	The signal generator backend is open. There is no capsule in the chain.
Levels arrive at exactly digital silence	macOS is denying microphone access. Since v0.1.2 the app asks on first launch; if you declined, allow it under System Settings → Privacy & Security → Microphone, or run `tccutil reset Microphone com.allansargeant.leqtion` and launch again. v0.1.1 and v0.2.0-beta.1 never asked at all — update.
LEQ looks short	Frames were dropped; the device bar will say so. Restart the measurement.
A-weighted reading disagrees with a real meter at HF	Expected at 44.1/48 kHz — about 1.1 dB by 10 kHz. Run at 96 kHz.
Transfer function trace breaks up	Coherence below the floor. That is the feature, not a fault — the measurement is not trustworthy there.
Transfer function looks confident but wrong	Check the reference is delay-compensated.
Generator clips at a level that looked safe	Level is dBFS **RMS**. Read the expected peak shown beside it.
ASIO build does nothing	It has never carried audio. Use WASAPI on Windows.

See also

- asio.md: the ASIO build flag and where it's at
- field-test.md: what a real-world check should cover
- README: the full DSP reasoning, and downloads